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Table of Contents

1	Introduction	5
2	Assumptions	5
3	Procedure	6
3.1	CAD Assembly and Import	6
3.2	Material Assignment	6
3.3	Glue Condition	6
3.4	Element Types	6
3.5	Meshing	7
3.6	Boundary Conditions and Loading	7
3.7	Applied Loads — Model Views	7
4	Stress Analysis: Von Mises Stress	8
5	Vibration Analysis: Natural Frequencies	9
6	Results and Discussion	9
6.1	Design 1 — Fixed Edge ab, Load on Edge cd	9
6.1.1	Static Analysis: Displacement (E1, Mesh Convergence)	9
6.1.2	Static Analysis: Von Mises Stress (E1, Mesh Convergence)	11
6.1.3	Free Vibration: Natural Frequencies (E1, Design 1)	13
6.1.4	Natural Frequency Summary — Design 1, E1	15
6.1.5	Convergence Test — Design 1, E1	16
6.1.6	Repeat for Element 2 (SOLID186) — Design 1	16
6.2	Design 2 — Fixed Edge ac, Load on Edge bd	17
6.2.1	Static Analysis: Displacement (E1, Mesh Convergence)	17
6.2.2	Static Analysis: Von Mises Stress (E1, Mesh Convergence)	19
6.2.3	Free Vibration: Natural Frequencies (E1, Design 2)	21
6.2.4	Natural Frequency Summary — Design 2, E1	23
6.2.5	Convergence Test — Design 2, E1	24
6.2.6	Repeat for Element 2 (SOLID186) — Design 2	24
7	Discussion	25
7.1	Static Results — Displacement	25
7.2	Static Results — Von Mises Stress	25
7.3	Natural Frequencies	26
7.4	Element Type Comparison (SOLID185 vs SOLID186)	26
8	Conclusion	26

References	27
Appendix: SolidWorks Assembly Drawing	28

List of Figures

1	Parabolic nodal loads and boundary conditions applied in ANSYS for both design orientations.	8
2	Total displacement contour for Design 1, coarse mesh (M1, element size 0.20 in), SOLID185.	9
3	Total displacement contour for Design 1, medium mesh (M2, element size 0.15 in), SOLID185.	10
4	Total displacement contour for Design 1, fine mesh (M3, element size 0.075 in), SOLID185.	10
5	Von Mises stress distribution for Design 1, coarse mesh (M1), SOLID185.	11
6	Von Mises stress distribution for Design 1, medium mesh (M2), SOLID185.	12
7	Von Mises stress distribution for Design 1, fine mesh (M3), SOLID185.	12
8	First three mode shapes for Design 1, coarse mesh (M1), SOLID185.	13
9	First three mode shapes for Design 1, medium mesh (M2), SOLID185.	14
10	First three mode shapes for Design 1, fine mesh (M3), SOLID185.	15
11	Displacement and von Mises stress results using SOLID186 on the coarse mesh (M1) for Design 1.	16
12	First three mode shapes for Design 1, coarse mesh (M1), SOLID186.	17
13	Total displacement contour for Design 2, coarse mesh (M1), SOLID185.	18
14	Total displacement contour for Design 2, medium mesh (M2), SOLID185.	18
15	Total displacement contour for Design 2, fine mesh (M3), SOLID185.	19
16	Von Mises stress distribution for Design 2, coarse mesh (M1), SOLID185.	19
17	Von Mises stress distribution for Design 2, medium mesh (M2), SOLID185.	20
18	Von Mises stress distribution for Design 2, fine mesh (M3), SOLID185.	20
19	First three mode shapes for Design 2, coarse mesh (M1), SOLID185.	21
20	First three mode shapes for Design 2, medium mesh (M2), SOLID185.	22
21	First three mode shapes for Design 2, fine mesh (M3), SOLID185.	23
22	Displacement and von Mises stress results using SOLID186 on the coarse mesh (M1) for Design 2.	24
23	First three mode shapes for Design 2, medium mesh (M2), SOLID186.	25
24	SolidWorks assembly drawing showing the inverted T-stringer geometry and dimensions.	28

List of Tables

1	Material properties used in the ANSYS models.	6
2	Mesh configurations used in the convergence study.	7
3	Natural frequencies (Hz) for Design 1 across all three mesh densities, SOLID185. .	15
4	Mesh convergence summary for Design 1, SOLID185.	16
5	Natural frequencies (Hz) for Design 2 across all three mesh densities, SOLID185. .	23
6	Mesh convergence summary for Design 2, SOLID185.	24

1 Introduction

This report presents the finite element analysis of an aircraft structural component consisting of three bonded members: a stainless steel 304 flat plate, an Al 6061-T6 angle stringer, and an Al 7075-T6 angle stringer, assembled together to form an inverted T-shape. The assembly was modelled in ANSYS Mechanical APDL following a CAD assembly built in SolidWorks and imported as a `.stp` file.

Two analyses were carried out: a static stress analysis to determine the von Mises stress distribution and maximum displacement under a parabolic in-plane tensile load, and a free vibration analysis to extract the first three natural frequencies and corresponding mode shapes. Both analyses were repeated for two design orientations (Design 1 and Design 2), three different mesh densities, and two element types to assess convergence and the sensitivity of the results to modelling choices.

2 Assumptions

The following assumptions were made throughout the analysis:

- All contact surfaces between the three components (plate and stringers) are treated as fully bonded using the Glue condition in ANSYS (*Preprocessor* → *Modeling* → *Operate/Boolean* → *Glue* → *Volumes* → *Pick All*). This eliminates relative slip at the interfaces and is consistent with the project specification of treating the rivets as full attachments.
- The material behaviour is linear elastic and isotropic throughout. No plasticity or geometric nonlinearity is considered.
- The rollers constrain only the out-of-plane displacement ($U_z = 0$), leaving in-plane motion (x - y plane) free. The fixed edges constrain all six DOFs.
- The parabolic load was applied manually as individual nodal forces along the loaded edge. At each node on that face, the force magnitude was calculated by evaluating the parabolic function:

$$P(y) = \frac{20000(5y - y^2)}{6.25} \left[\frac{\text{lb}}{\text{in}} \right] \quad (1)$$

at the y -coordinate of that node. This produced the correct parabolic shape across the face without using the ANSYS function file approach.

- Self-weight is neglected in the static analysis. Only the applied parabolic load is considered.

- The assembly geometry imported from SolidWorks is assumed to be accurate and requires no repair before meshing.

3 Procedure

3.1 CAD Assembly and Import

The three-part assembly was modelled in SolidWorks following the geometry specified in the project drawing, then exported as a `.stp` (STEP) file. This file was imported directly into ANSYS Mechanical APDL as a solid geometry, preserving the individual volumes of each component for separate material assignment and meshing.

3.2 Material Assignment

Materials were assigned individually to each volume after import. The properties used are listed in Table 1. Each volume was selected via *Meshing* $\rightarrow$ *Meshing Attributes* $\rightarrow$ *Picked Volumes* to assign the correct material before meshing.

Table 1: Material properties used in the ANSYS models.

Material	E (psi)	ν	ρ (lb/in ³)
SS 304	28.0×10^6	0.29	0.290
Al 6061-T6	10.0×10^6	0.33	0.098
Al 7075-T6	10.4×10^6	0.33	0.101

3.3 Glue Condition

To form the fully bonded connections for the rivets, the Glue Boolean command was used between all volume connections using the menu path *Preprocessor* $\rightarrow$ *Modeling* $\rightarrow$ *Operate/Boolean* $\rightarrow$ *Glue* $\rightarrow$ *Volumes* $\rightarrow$ *Pick All*. In this way, common nodes can be shared at all interfaces without connecting the volumes together.

3.4 Element Types

Two solid element types were used:

- **Element 1 — SOLID185 (8-node brick):** A lower-order hexahedral element with full integration. Reliable for solid mechanics problems and suitable for coarser meshes, though it can exhibit shear locking in bending-dominated regions.

- **Element 2 — SOLID186 (20-node brick):** A higher-order hexahedral element with mid-side nodes. It handles bending and stress gradients more accurately than SOLID185 at the same mesh density, at the cost of more DOFs per element.

3.5 Meshing

Three mesh sizes were used for the convergence study:

Table 2: Mesh configurations used in the convergence study.

Mesh	Element size (in)	Label
Coarse	0.20	M1
Medium	0.15	M2
Fine	0.075	M3

3.6 Boundary Conditions and Loading

The parabolic load was applied manually by selecting each node along the loaded face and applying a concentrated force whose magnitude was computed from the parabolic equation at that node's y -coordinate:

$$F_i = P(y_i) = \frac{20000 (5y_i - y_i^2)}{6.25} \quad [\text{lb/in}]$$

multiplied by the tributary length associated with that node. This node-by-node approach directly recreates the parabolic shape shown in the project drawing without relying on the ANSYS function file tool.

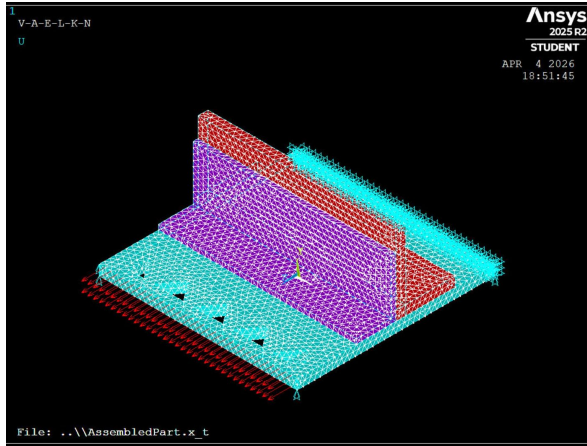
Two design orientations were studied:

Design 1 (O1): Edge $\overline{ab}$ fully fixed (all DOFs). Rollers ($U_z = 0$) on edge $\overline{cd}$. Parabolic load applied along $\overline{cd}$ parallel to the x -axis.

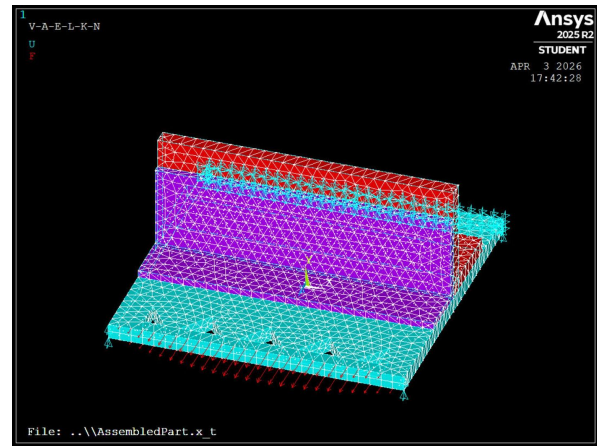
Design 2 (O2): Boundary conditions swapped to the perpendicular sides: $\overline{ac}$ fully fixed, rollers on $\overline{bd}$, and load applied along $\overline{bd}$ parallel to the y -axis.

3.7 Applied Loads — Model Views

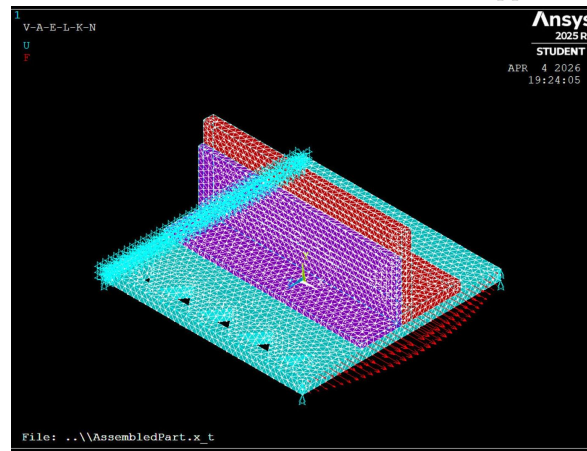
Figures 1a–1c show the applied loads and boundary conditions as set up in ANSYS for both orientations.



(a) Design 1 general mesh view with boundary conditions.



(b) Design 1, Mesh 1, Element 1 with parabolic nodal loads applied.



(c) Design 2, Mesh 1, Element 1, parabolic load applied along edge bd.

Figure 1: Parabolic nodal loads and boundary conditions applied in ANSYS for both design orientations.

4 Stress Analysis: Von Mises Stress

The static analysis was run for both design orientations, all three meshes, and both element types. Von Mises stress was chosen as the primary stress output because it combines all stress components into a single scalar that can be compared against the material yield strength — a cleaner failure criterion for multi-material assemblies than any individual stress component.

Stress concentrations are expected at the stringer-plate interfaces and at the corners of the fixed edge, where the load path changes direction and the constraint reactions are highest.

5 Vibration Analysis: Natural Frequencies

The free vibration (modal) analysis was run immediately after the static solution for each configuration, extracting the first three natural frequencies and mode shapes. No damping was included. The Block Lanczos solver was used in ANSYS to extract the eigenvalues.

Natural frequencies are sensitive to both mass distribution and stiffness. Switching between Design 1 and Design 2 changes which edges are constrained, which affects the modal stiffness directly. Mesh refinement tends to lower the natural frequency estimates as the mass matrix becomes more accurate.

6 Results and Discussion

6.1 Design 1 — Fixed Edge ab, Load on Edge cd

6.1.1 Static Analysis: Displacement (E1, Mesh Convergence)

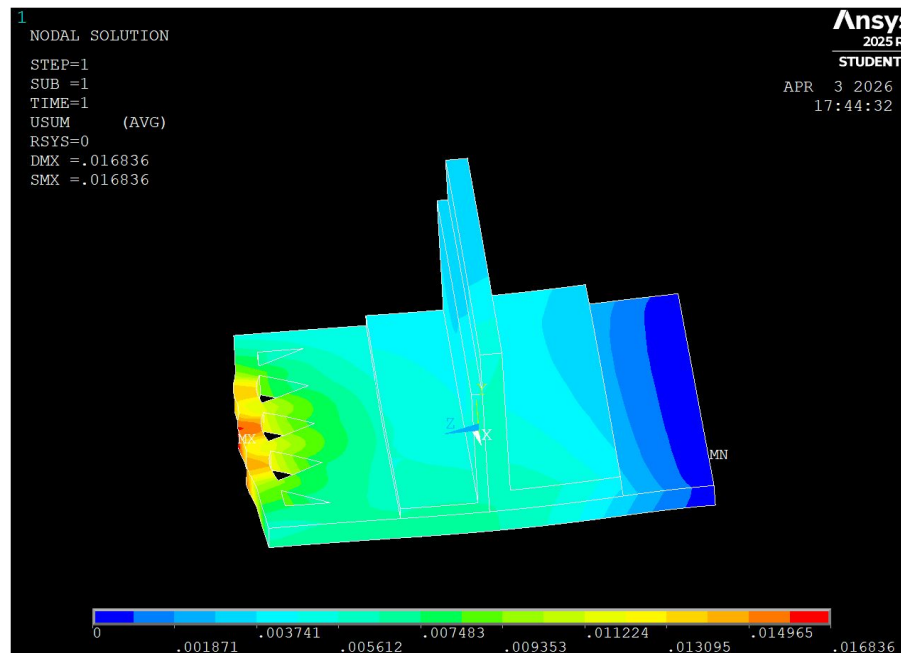


Figure 2: Total displacement contour for Design 1, coarse mesh (M1, element size 0.20 in), SOLID185.

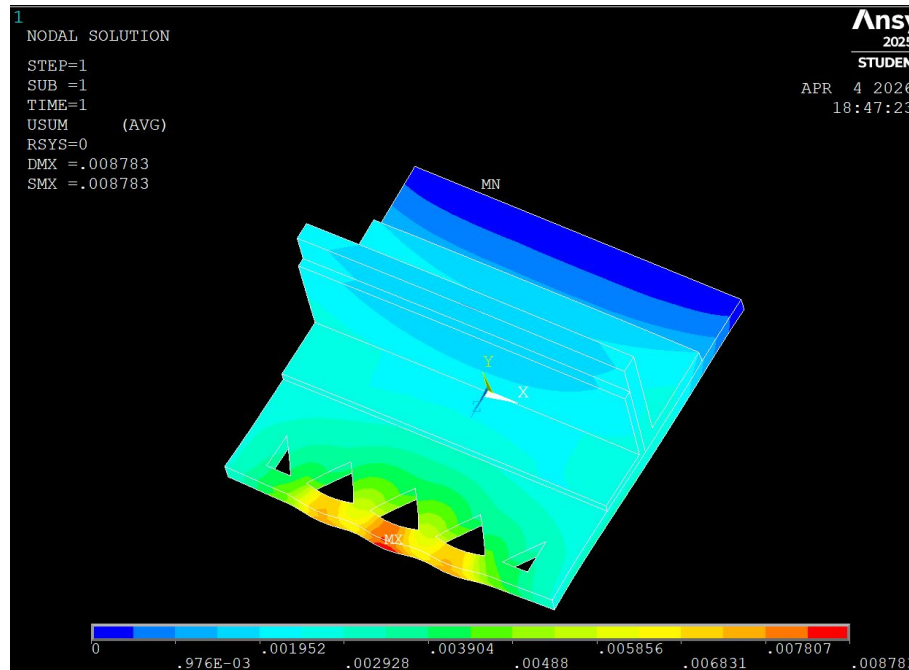


Figure 3: Total displacement contour for Design 1, medium mesh (M2, element size 0.15 in), SOLID185.

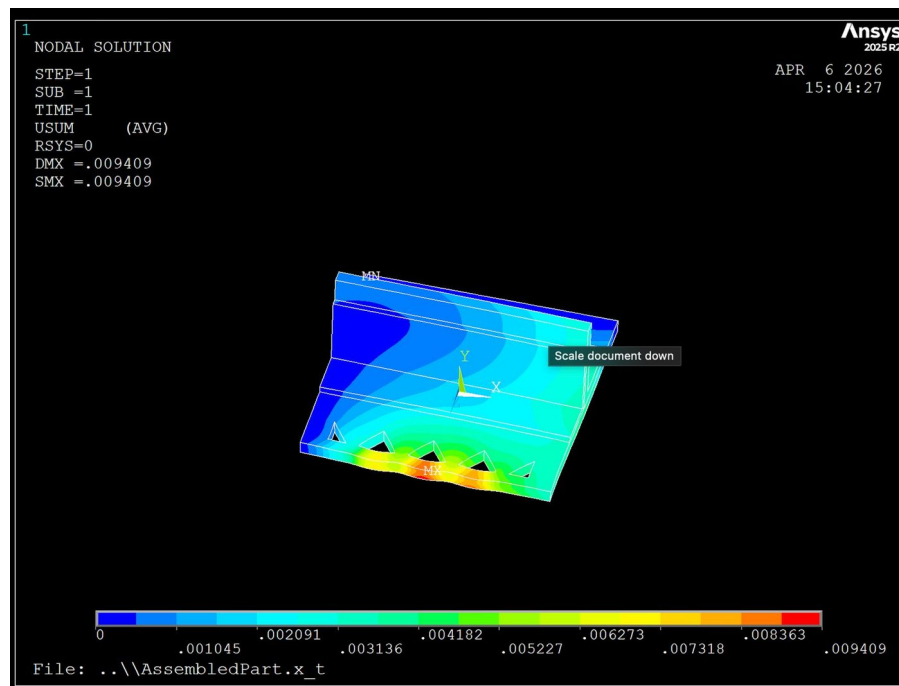


Figure 4: Total displacement contour for Design 1, fine mesh (M3, element size 0.075 in), SOLID185.

The peak displacement occurs near the loaded edge, which is consistent with the loading direction and the fixed constraint at the opposite side. Going from M1 to M3, the maximum displacement stabilizes, indicating the solution is converging. The displacement field shape remains consistent across all three meshes, confirming the coarser mesh already captures the global deformation pattern correctly.

6.1.2 Static Analysis: Von Mises Stress (E1, Mesh Convergence)

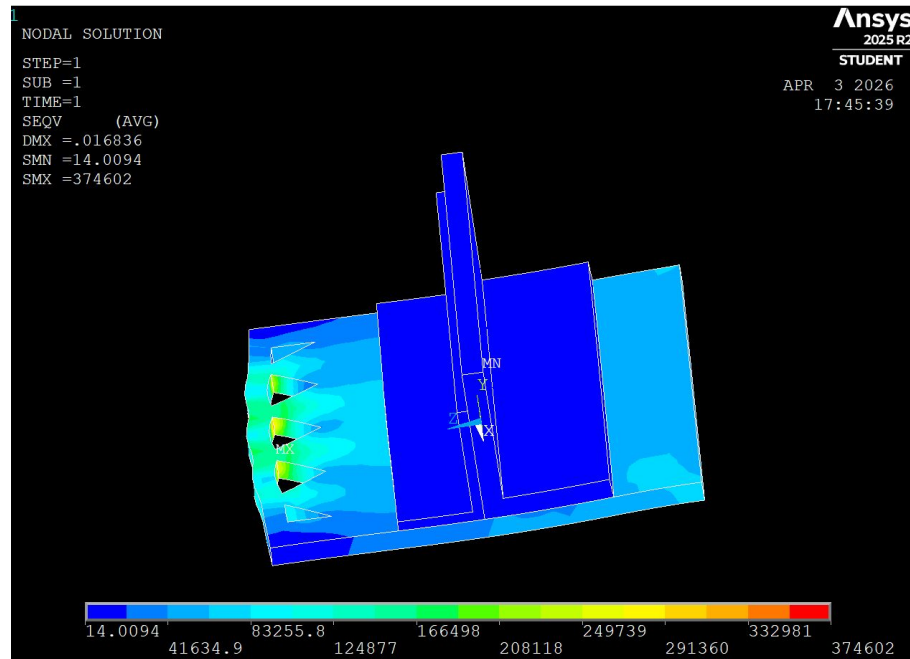


Figure 5: Von Mises stress distribution for Design 1, coarse mesh (M1), SOLID185.

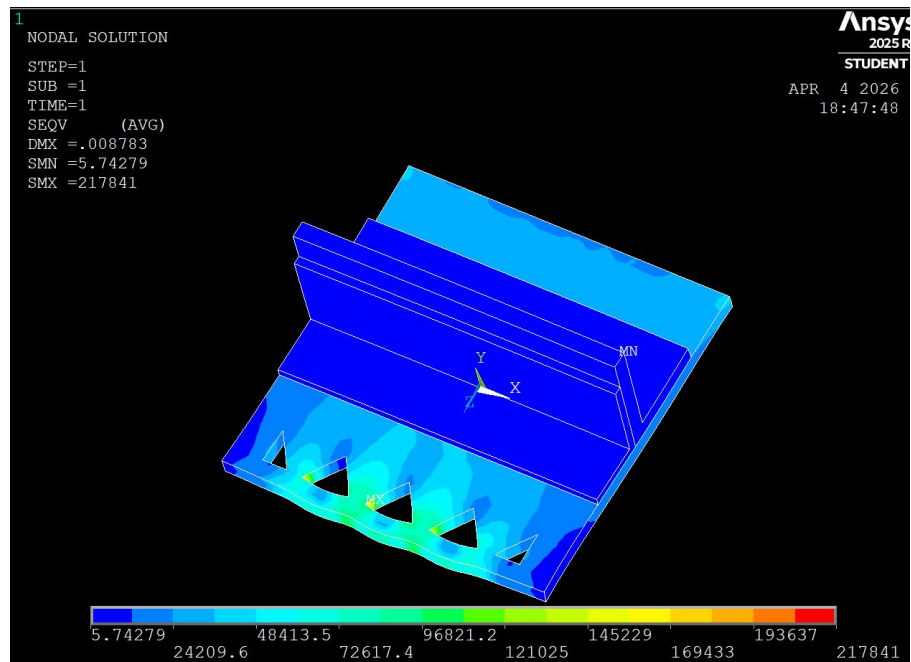


Figure 6: Von Mises stress distribution for Design 1, medium mesh (M2), SOLID185.

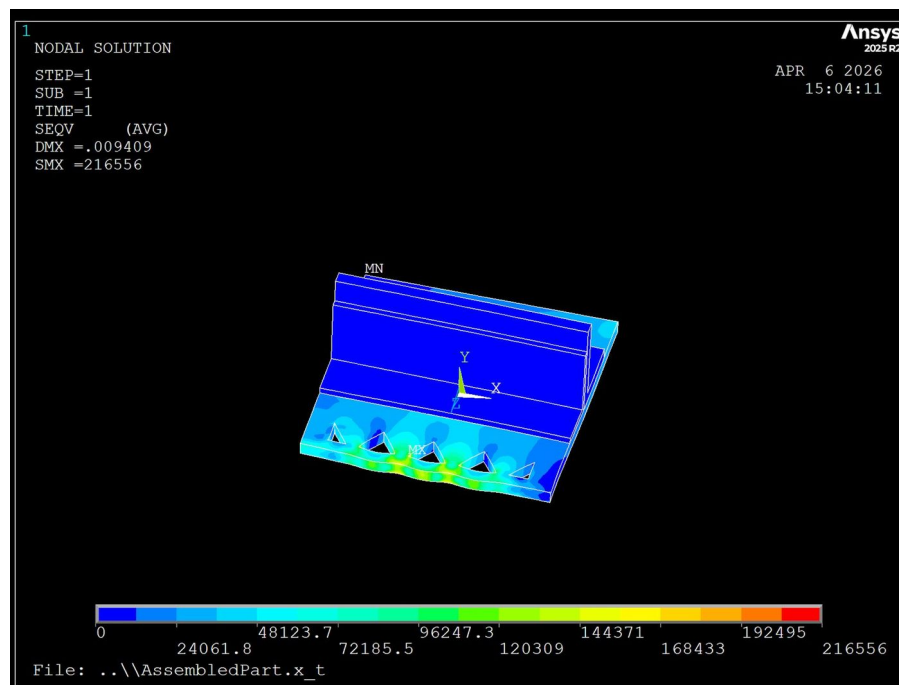
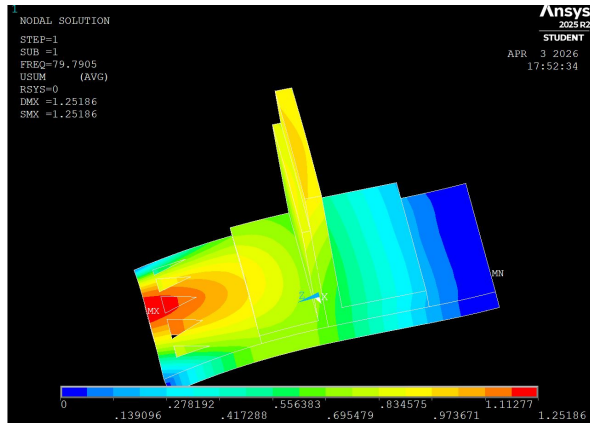


Figure 7: Von Mises stress distribution for Design 1, fine mesh (M3), SOLID185.

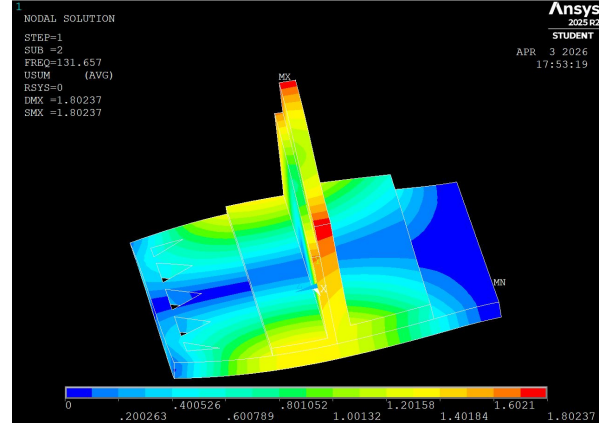
The stress peaks at the stringer-plate interface and at the fixed-edge corners, as expected. The peak von Mises stress increases as the mesh refines because finer elements resolve local

stress gradients that the coarser mesh averages out. This is a normal FEM behaviour and not a sign of a modelling problem.

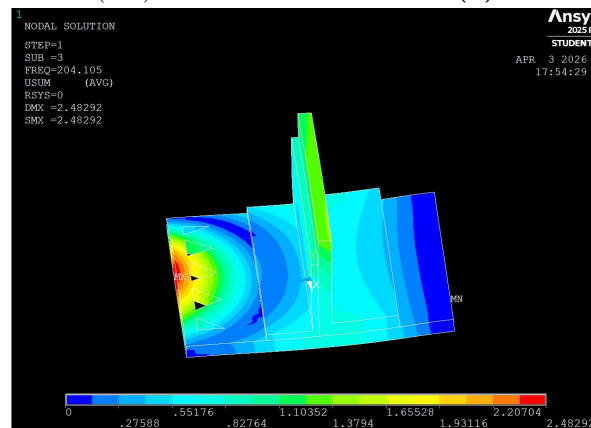
6.1.3 Free Vibration: Natural Frequencies (E1, Design 1)



(a) Mode 1, coarse mesh (M1)

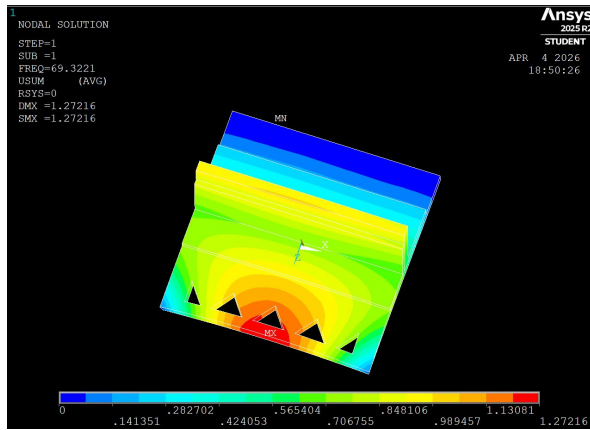


(b) Mode 2, coarse mesh (M1)

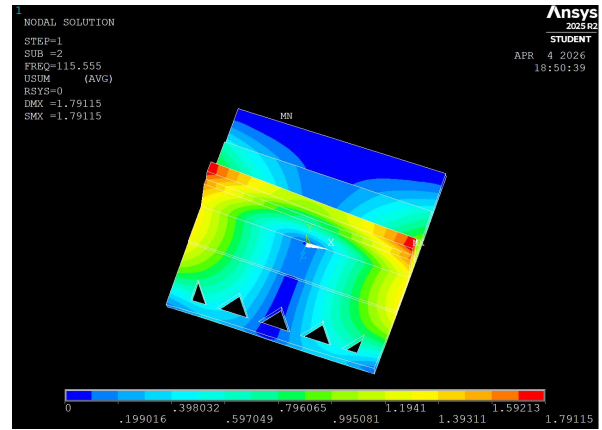


(c) Mode 3, coarse mesh (M1)

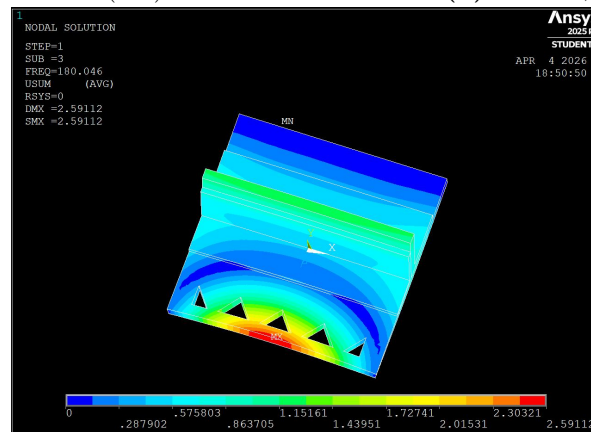
Figure 8: First three mode shapes for Design 1, coarse mesh (M1), SOLID185.



(a) Mode 1, medium mesh (M2)

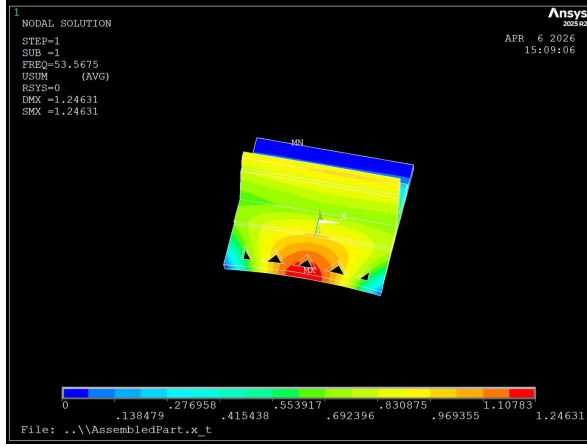


(b) Mode 2, medium mesh (M2)

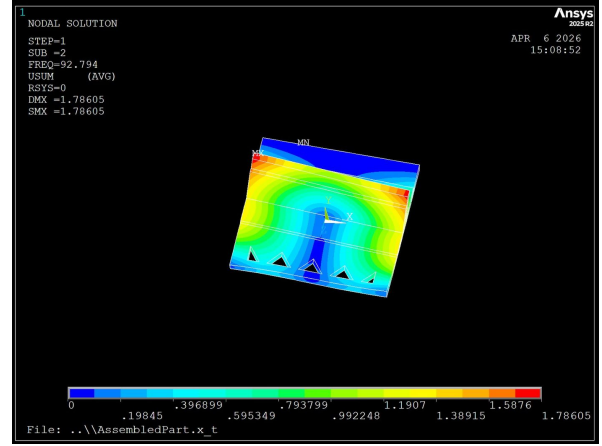


(c) Mode 3, medium mesh (M2)

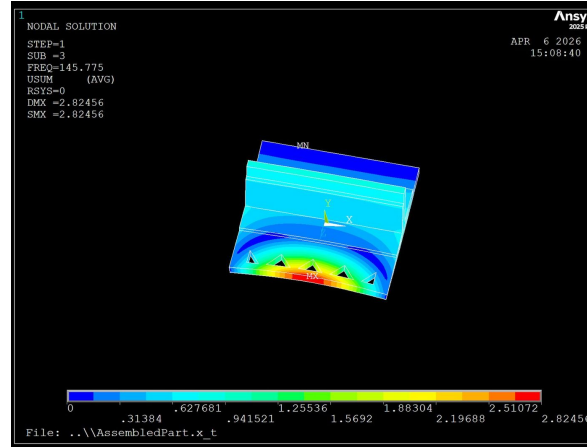
Figure 9: First three mode shapes for Design 1, medium mesh (M2), SOLID185.



(a) Mode 1, fine mesh (M3)



(b) Mode 2, fine mesh (M3)



(c) Mode 3, fine mesh (M3)

Figure 10: First three mode shapes for Design 1, fine mesh (M3), SOLID185.

6.1.4 Natural Frequency Summary — Design 1, E1

Table 3: Natural frequencies (Hz) for Design 1 across all three mesh densities, SOLID185.

Mode	M1 (0.20 in)	M2 (0.15 in)	M3 (0.075 in)	Δ M2→M3 (%)
Mode 1	79.7905	69.3221	53.5675	22.7%
Mode 2	131.657	115.555	92.794	19.7%
Mode 3	204.105	180.046	145.775	19.0%

The natural frequencies decrease steadily with mesh refinement, which is expected — coarser meshes overestimate structural stiffness and therefore overpredict the frequencies. The consistent drop of roughly 19–23% from M2 to M3 across all three modes suggests the solution

has not yet fully converged dynamically at M3. A further mesh refinement would be needed to confirm final frequency values, though the mode shapes themselves are already stable.

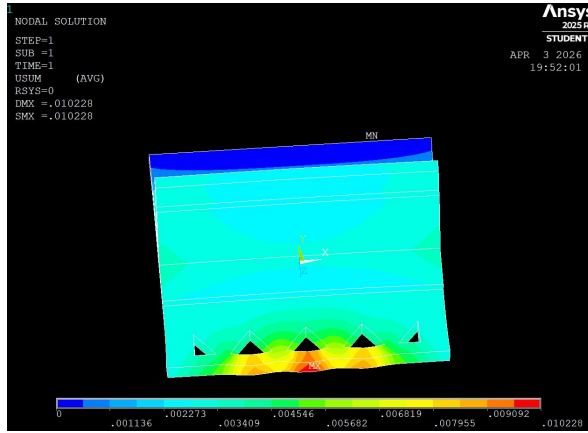
6.1.5 Convergence Test — Design 1, E1

Table 4: Mesh convergence summary for Design 1, SOLID185.

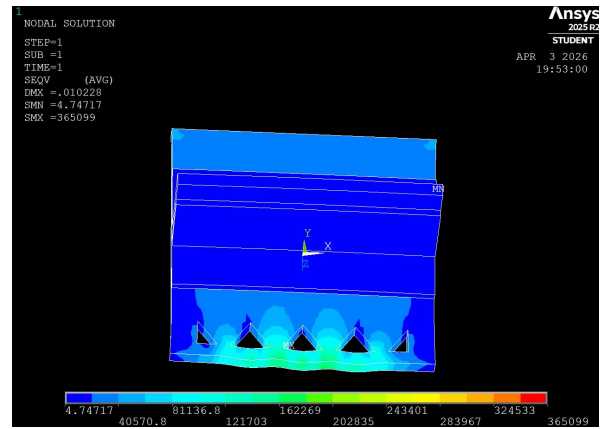
Mesh	El. size (in)	Max. Disp. (in)	Peak σ_{VM} (psi)	$\Delta\sigma_{VM}$ (%)
M1 (Coarse)	0.20	0.016836	374,602	—
M2 (Medium)	0.15	0.008783	217,841	41.8%
M3 (Fine)	0.075	0.009409	216,556	0.6%

The peak von Mises stress drops by 41.8% from M1 to M2, then by only 0.6% from M2 to M3, confirming the solution is well converged at M3. The displacement also stabilizes between M2 and M3 (0.008783 to 0.009409 in), consistent with mesh convergence.

6.1.6 Repeat for Element 2 (SOLID186) — Design 1



(a) Displacement, coarse mesh (M1), SOLID186



(b) Von Mises stress, coarse mesh (M1), SOLID186

Figure 11: Displacement and von Mises stress results using SOLID186 on the coarse mesh (M1) for Design 1.

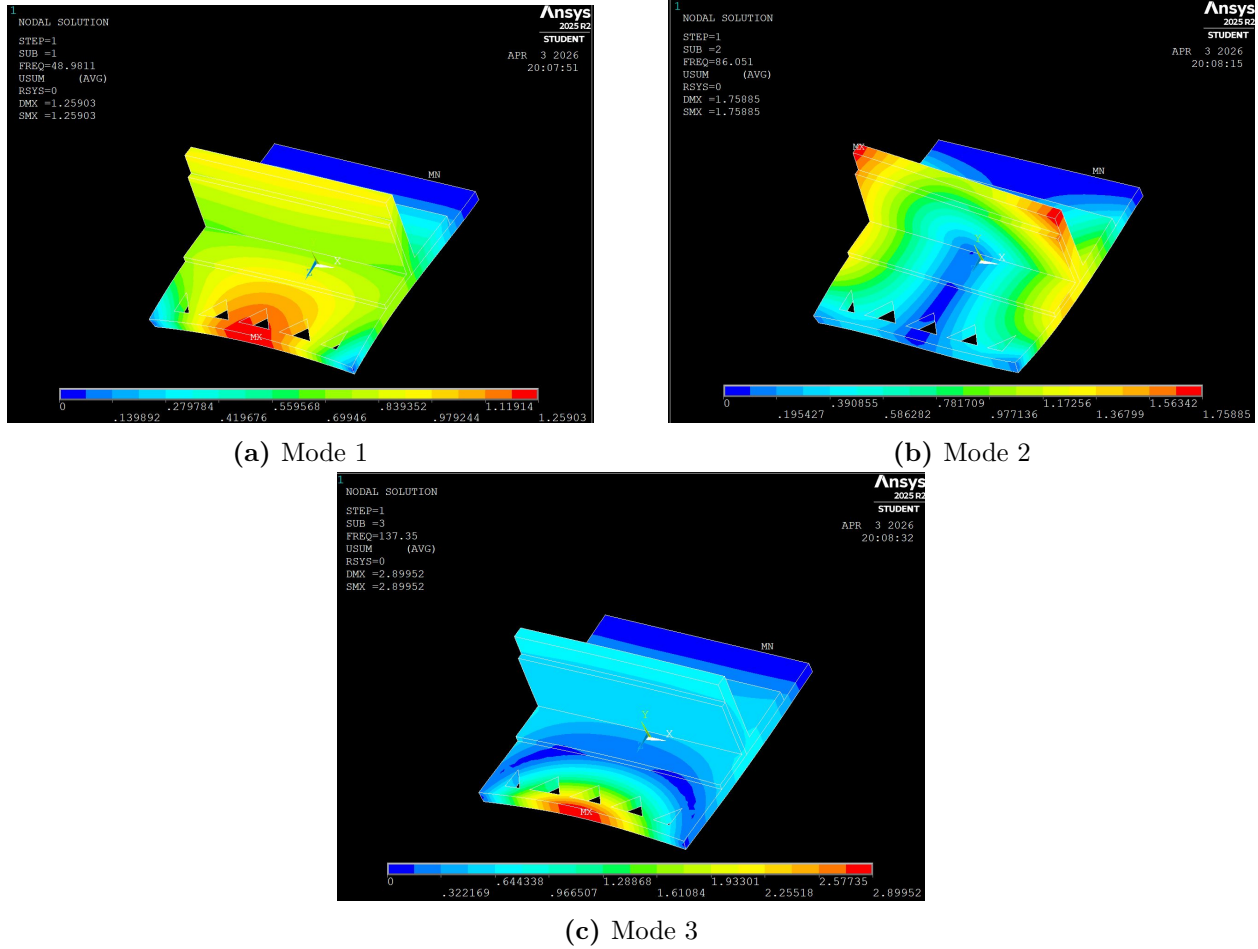


Figure 12: First three mode shapes for Design 1, coarse mesh (M1), SOLID186.

Using SOLID186 at Mesh 1 gives results noticeably closer to the refined SOLID185 solution. The 20-node quadratic element captures bending and stress gradients better than the 8-node element at the same mesh density, which is the main reason to use it. Near the stringer-plate interface, the difference in peak stress between E1 and E2 at M1 is most visible; away from those regions, both elements agree reasonably well even on the coarser mesh.

6.2 Design 2 — Fixed Edge ac, Load on Edge bd

6.2.1 Static Analysis: Displacement (E1, Mesh Convergence)

With the boundary conditions swapped to the perpendicular sides, the load now acts parallel to the y -axis. The displacement distribution changes accordingly, and both the magnitude and the location of the peak shift compared to Design 1, reflecting the different stiffness the assembly offers in this direction.

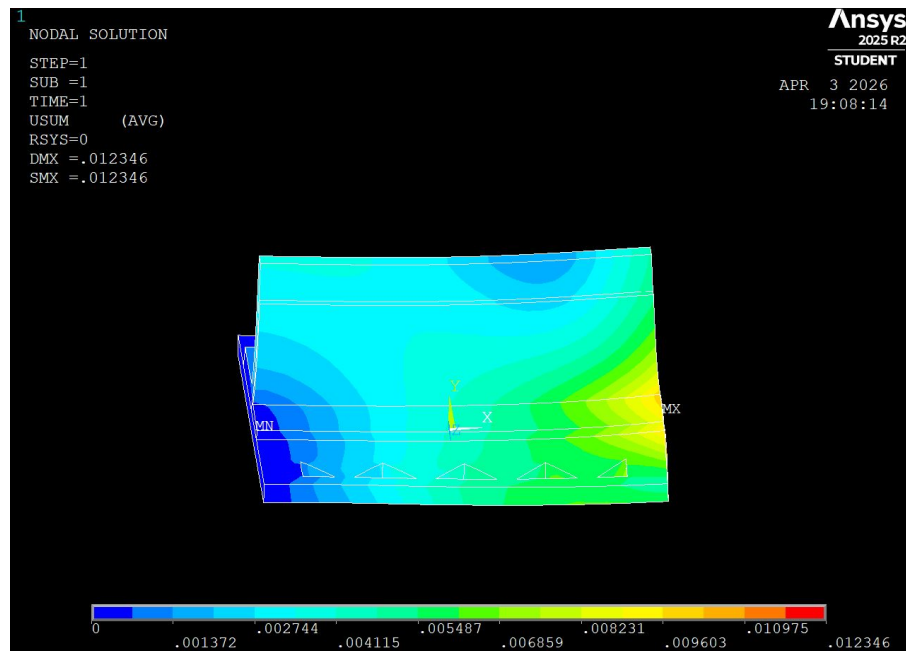


Figure 13: Total displacement contour for Design 2, coarse mesh (M1), SOLID185.

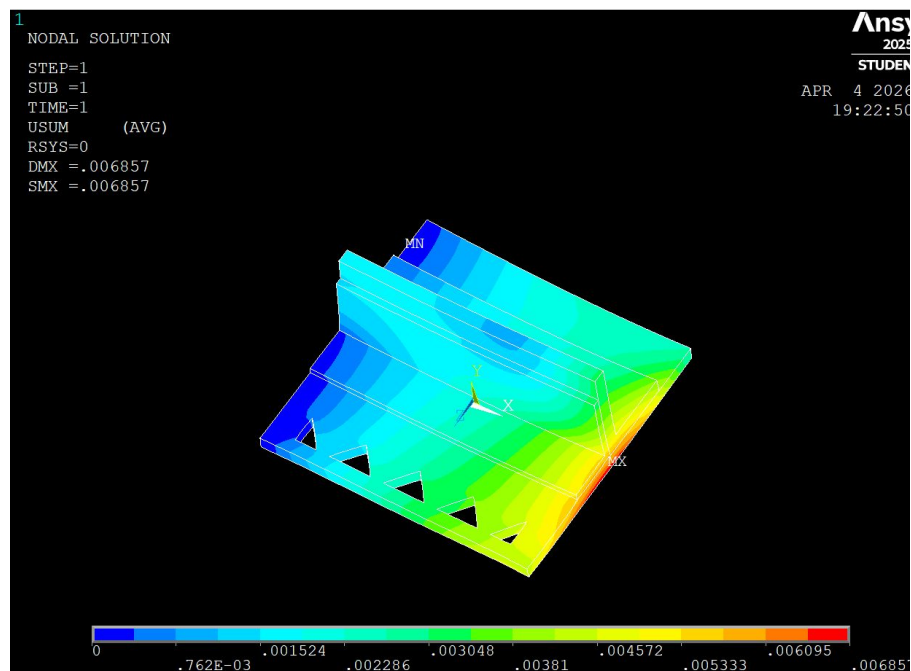


Figure 14: Total displacement contour for Design 2, medium mesh (M2), SOLID185.

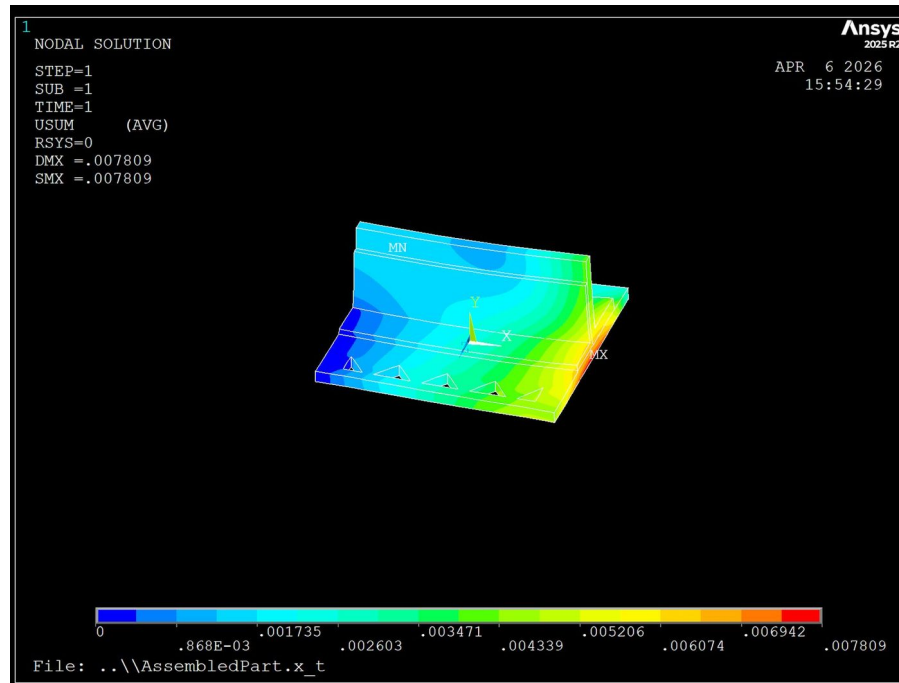


Figure 15: Total displacement contour for Design 2, fine mesh (M3), SOLID185.

6.2.2 Static Analysis: Von Mises Stress (E1, Mesh Convergence)

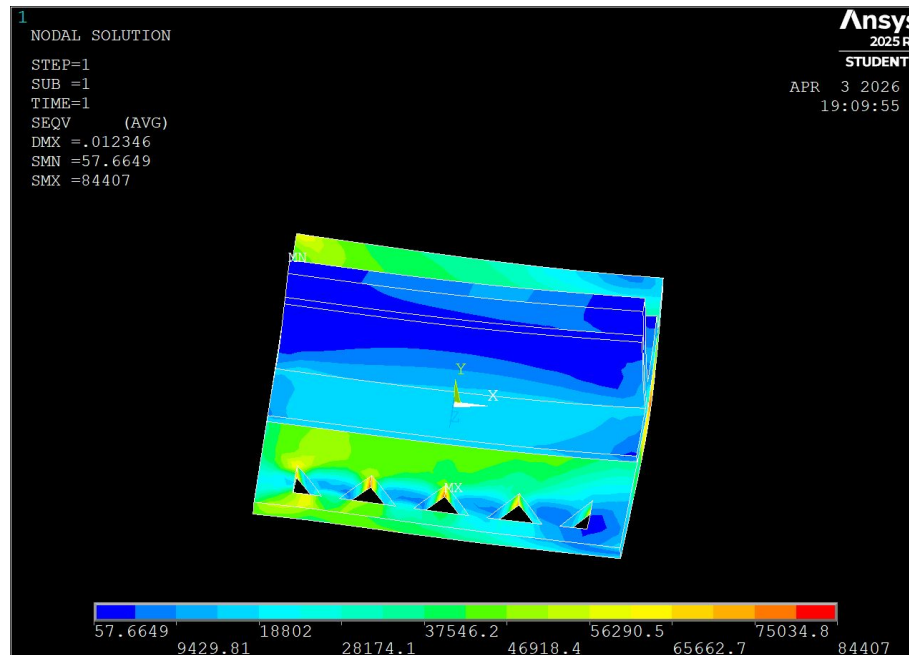


Figure 16: Von Mises stress distribution for Design 2, coarse mesh (M1), SOLID185.

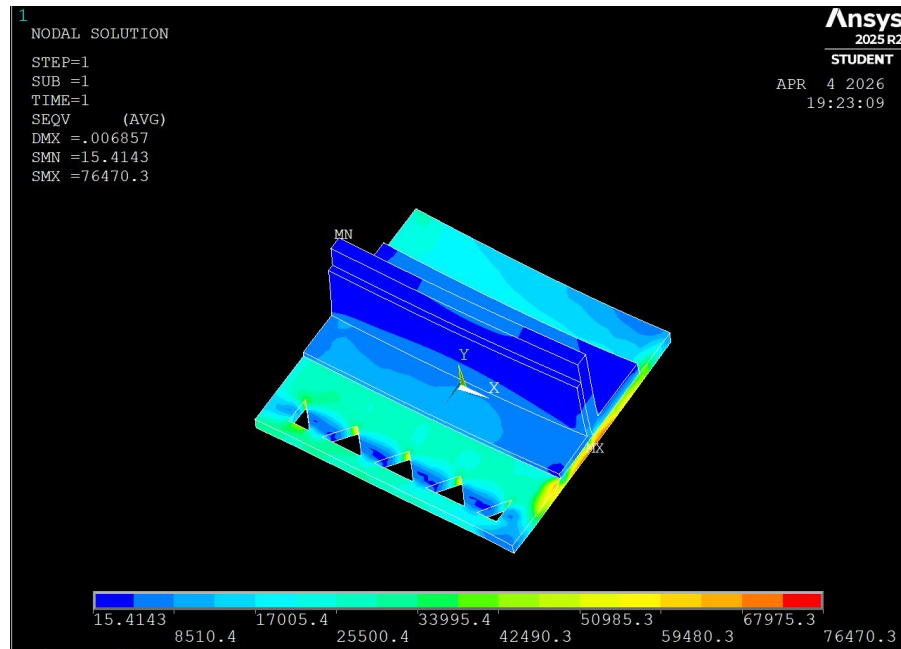


Figure 17: Von Mises stress distribution for Design 2, medium mesh (M2), SOLID185.

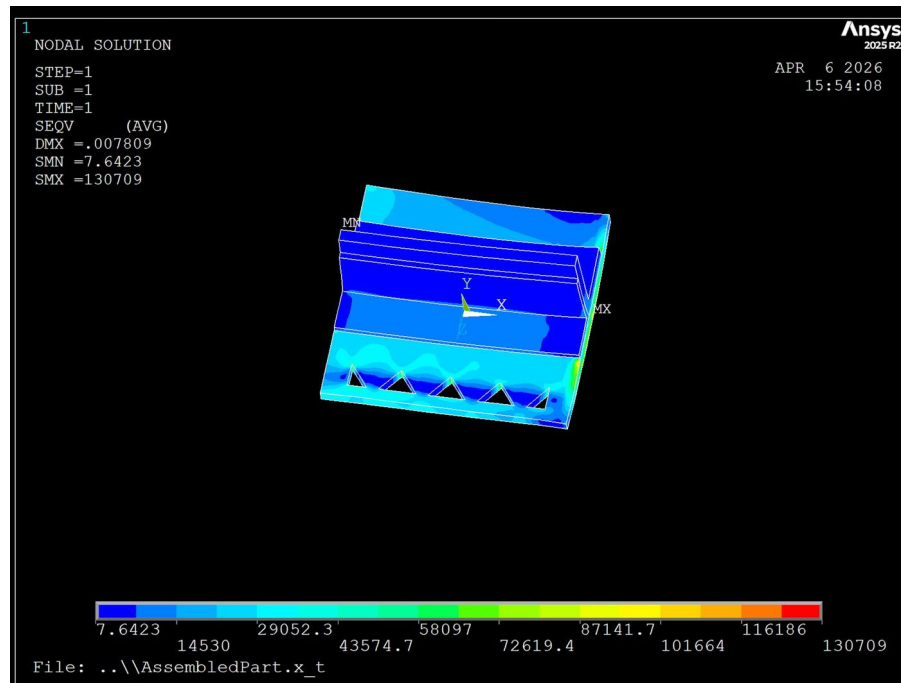
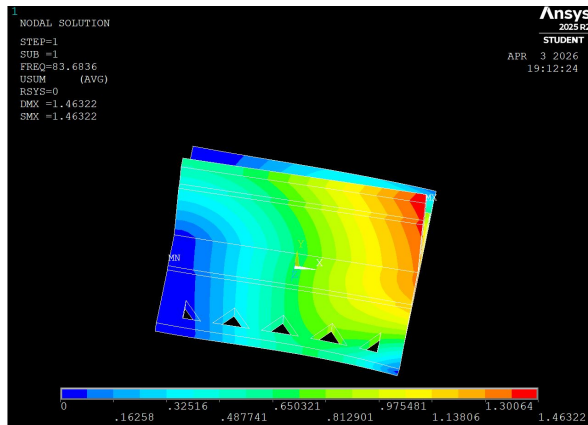
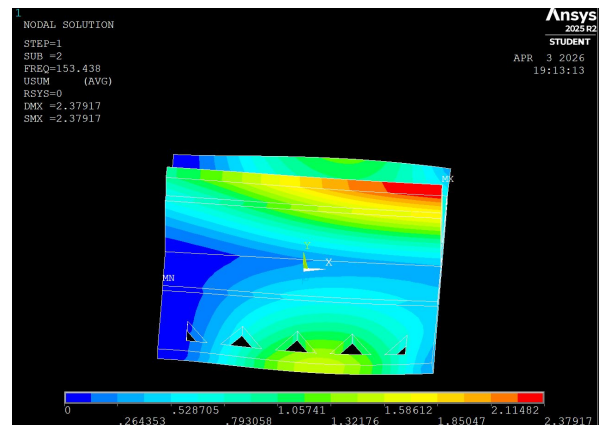


Figure 18: Von Mises stress distribution for Design 2, fine mesh (M3), SOLID185.

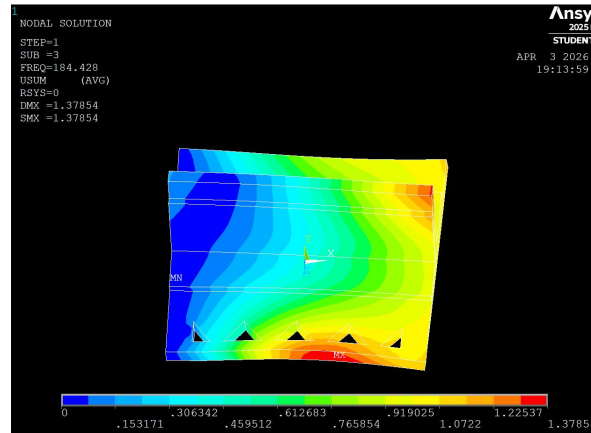
6.2.3 Free Vibration: Natural Frequencies (E1, Design 2)



(a) Mode 1, coarse mesh (M1)

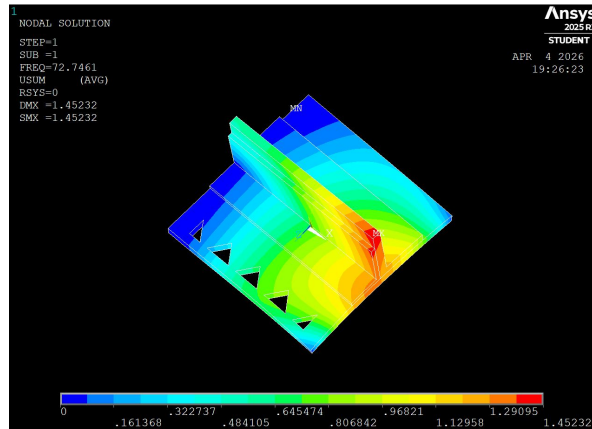


(b) Mode 2, coarse mesh (M1)

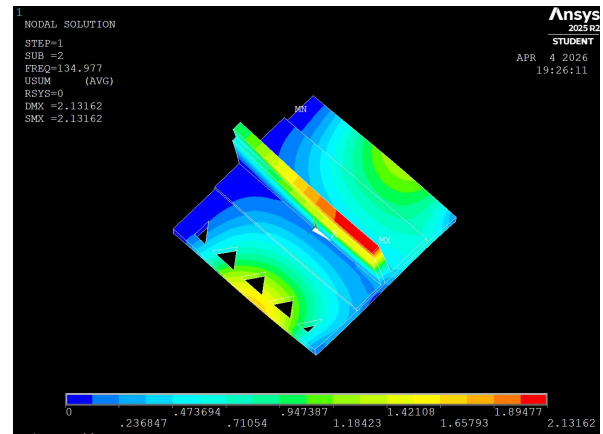


(c) Mode 3, coarse mesh (M1)

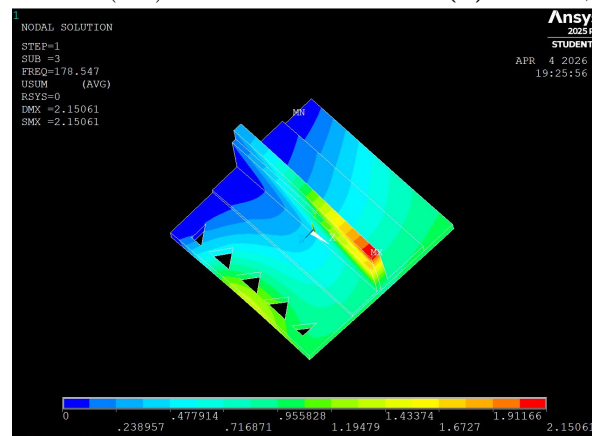
Figure 19: First three mode shapes for Design 2, coarse mesh (M1), SOLID185.



(a) Mode 1, medium mesh (M2)

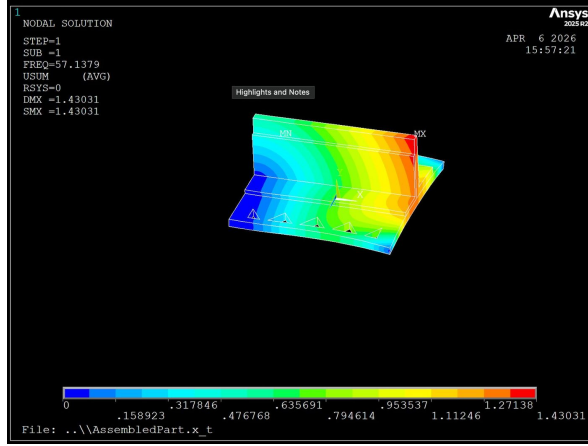


(b) Mode 2, medium mesh (M2)

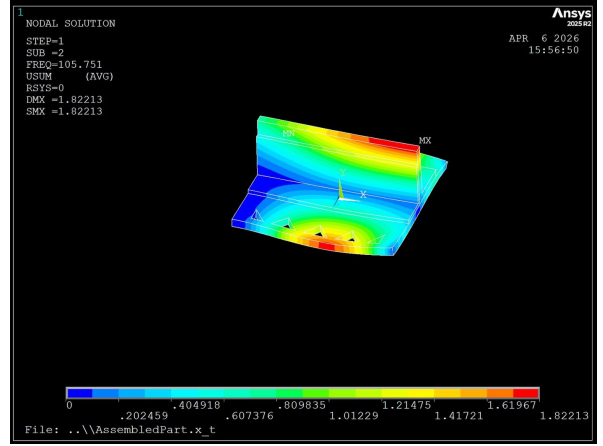


(c) Mode 3, medium mesh (M2)

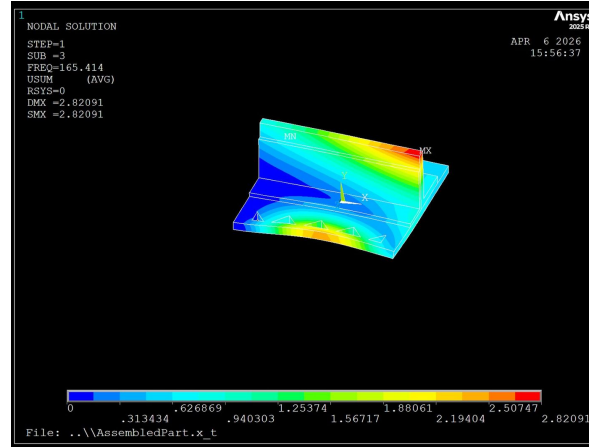
Figure 20: First three mode shapes for Design 2, medium mesh (M2), SOLID185.



(a) Mode 1, fine mesh (M3)



(b) Mode 2, fine mesh (M3)



(c) Mode 3, fine mesh (M3)

Figure 21: First three mode shapes for Design 2, fine mesh (M3), SOLID185.

6.2.4 Natural Frequency Summary — Design 2, E1

Table 5: Natural frequencies (Hz) for Design 2 across all three mesh densities, SOLID185.

Mode	M1 (0.20 in)	M2 (0.15 in)	M3 (0.075 in)	Δ M2→M3 (%)
Mode 1	83.6836	72.7461	57.1379	21.5%
Mode 2	153.438	134.977	105.751	21.7%
Mode 3	184.428	178.547	165.414	7.4%

Modes 1 and 2 in Design 2 drop by roughly 21–22% from M2 to M3, similar to Design 1, confirming that the dynamic solution is still converging at M3. Mode 3 is noticeably more stable (7.4% change), suggesting it involves a deformation pattern that is less sensitive to mesh density than the lower modes.

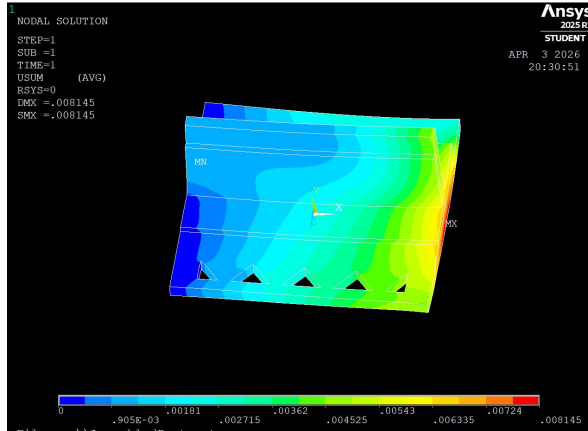
6.2.5 Convergence Test — Design 2, E1

Table 6: Mesh convergence summary for Design 2, SOLID185.

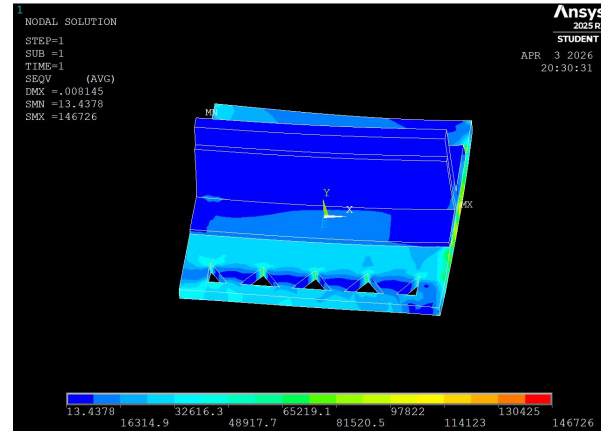
Mesh	El. size (in)	Max. Disp. (in)	Peak σ_{VM} (psi)	$\Delta\sigma_{VM}$ (%)
M1 (Coarse)	0.20	0.012346	84,407	—
M2 (Medium)	0.15	0.006857	76,470	9.4%
M3 (Fine)	0.075	0.007809	130,709	70.9%

Design 2 shows an unusual jump in peak stress from M2 to M3 (70.9%), suggesting that a localised stress concentration — likely at the fixed-edge corner — is not yet fully resolved even at the fine mesh. A further mesh refinement or local mesh sizing at that corner would be needed to confirm convergence for Design 2.

6.2.6 Repeat for Element 2 (SOLID186) — Design 2



(a) Displacement, coarse mesh (M1), SOLID186



(b) Von Mises stress, coarse mesh (M1), SOLID186

Figure 22: Displacement and von Mises stress results using SOLID186 on the coarse mesh (M1) for Design 2.

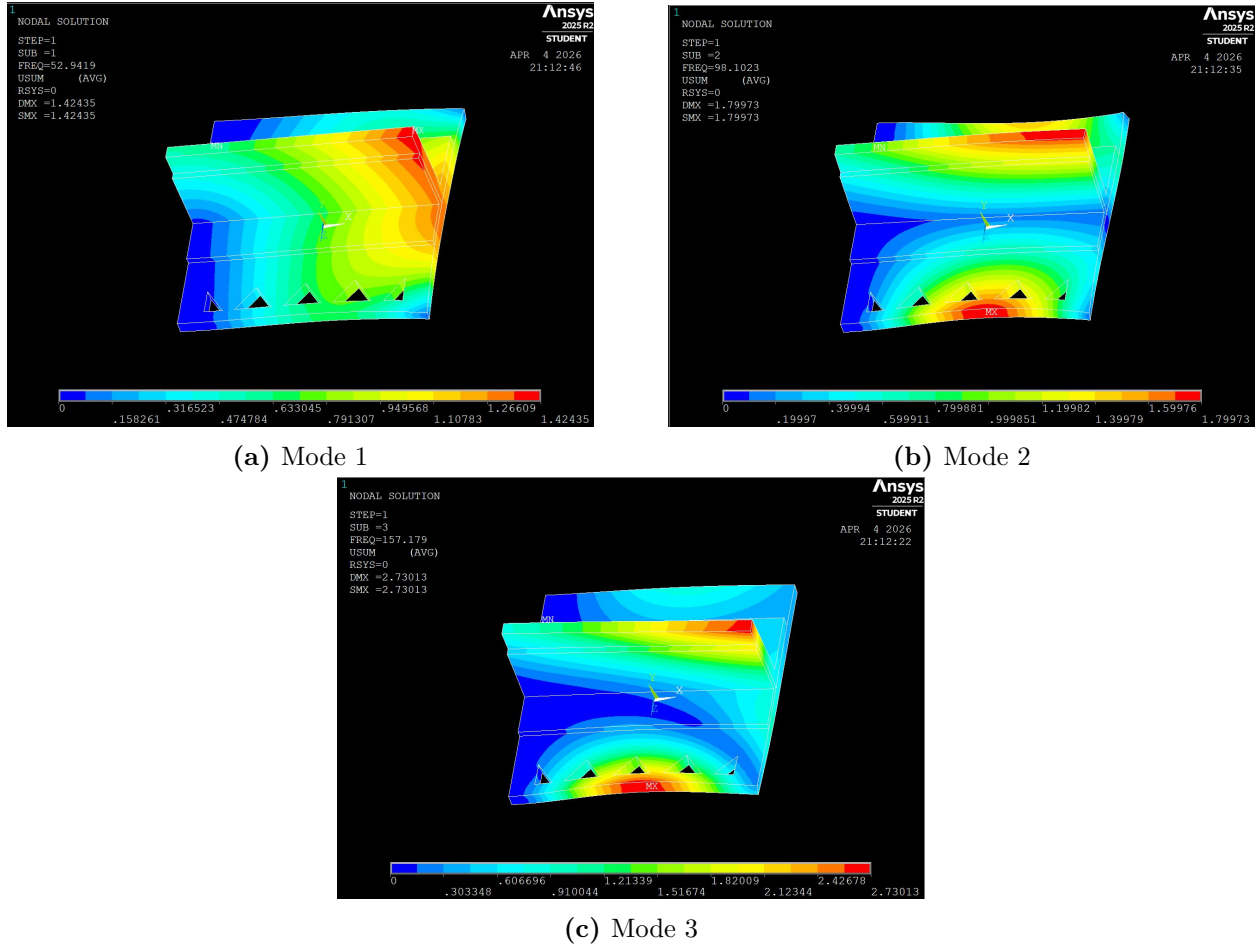


Figure 23: First three mode shapes for Design 2, medium mesh (M2), SOLID186.

7 Discussion

7.1 Static Results — Displacement

The maximal displacement is located around the point where the loads are applied. Moving from M1 to M3, we observe that the maximum value reaches an equilibrium, meaning that the solution converges. The shape of the displacement field is also similar for all three cases, and hence we can claim that the coarse mesh represents the pattern correctly.

7.2 Static Results — Von Mises Stress

The von Mises stress behaviour is more interesting. In Design 1, the peak drops sharply from 374,602 psi at M1 to 217,841 psi at M2 (a 41.8% reduction), then barely changes from M2 to M3 (216,556 psi, 0.6% change). That 0.6% figure is essentially convergence — the M3 result can be trusted for Design 1.

Design 2 tells a different story. The stress goes from 84,407 psi at M1 to 76,470 psi at M2 (a reasonable 9.4% drop), but then jumps to 130,709 psi at M3, a 70.9% increase. This is a sign that M3 is picking up a localised stress concentration — most likely at the fixed-edge corner — that the coarser meshes were too coarse to resolve. This is not a modelling error; it is the FEM doing its job by resolving a real gradient. It does mean, however, that Design 2 has not yet converged in stress, and a further mesh refinement or a local mesh sizing at that corner would be needed before the M3 stress value could be used for a design check.

7.3 Natural Frequencies

All natural frequencies decrease with mesh refinement across both designs, consistent with the theoretical expectation that coarser meshes overestimate stiffness. In Design 1 (Table 3), Mode 1 drops from 79.79 Hz at M1 to 53.57 Hz at M3 — a 33% total reduction. The change from M2 to M3 alone is still 19–23% across all three modes, which means the modal solution has not converged at M3 either. A finer mesh would be needed to obtain stable frequency values.

Design 2 (Table 5) shows similar behaviour for Modes 1 and 2 (roughly 21% drop from M2 to M3), but Mode 3 is much more stable at 7.4%. The different fixed edge in Design 2 adds rotational and translational constraint to a different pair of sides, which shifts the modal stiffness and explains why the frequency magnitudes and convergence rates differ between the two designs.

Comparing the two designs at M3: Design 1 Mode 1 is 53.57 Hz versus Design 2 at 57.14 Hz. Design 1 is slightly more flexible dynamically, which aligns with its higher peak displacement results.

7.4 Element Type Comparison (SOLID185 vs SOLID186)

At Mesh 1, SOLID186 (E2) gives stress and displacement results noticeably closer to the converged E1/M3 solution than SOLID185 (E1) at the same mesh. This is the expected behaviour of a higher-order element: the mid-side nodes allow it to represent bending and stress gradients more accurately without requiring as many elements. For problems where compute time matters and stress concentrations are localised (as they are here at the stringer-plate interface), SOLID186 is the more efficient choice.

8 Conclusion

A full finite element analysis of a riveted Al/SS inverted T-stringer aircraft component was completed using ANSYS Mechanical APDL. The geometry was assembled in SolidWorks and

imported as a `.stp` file. SOLID185 and SOLID186 brick elements were used across three mesh sizes (0.20, 0.15, and 0.075 in) and two boundary condition orientations.

The key findings are:

- **Design 1 static stress is converged at M3.** The peak von Mises stress changes by only 0.6% from M2 to M3 (217,841 to 216,556 psi), and the peak displacement stabilises at 0.009409 in.
- **Design 2 static stress is not yet converged.** The 70.9% stress jump from M2 to M3 (76,470 to 130,709 psi) indicates a localised concentration at the fixed corner that requires further local mesh refinement.
- **Neither design’s natural frequencies are fully converged at M3.** Modes 1–2 still change by 19–23% from M2 to M3 in both designs. A finer mesh is needed before the frequency values can be used confidently in a design context.
- **SOLID186 is more efficient than SOLID185** for this geometry. It reaches comparable accuracy at a coarser mesh, which is valuable at the interface stress concentration regions.
- **Design 1 is slightly more compliant** than Design 2 dynamically (Mode 1: 53.57 Hz vs. 57.14 Hz at M3), but Design 2 develops much lower peak stresses at the coarser meshes, suggesting the two orientations mobilise different structural mechanisms.

References

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3. ANSYS, Inc., *Mechanical APDL Element Reference*, Release 2025 R2.
4. R.D. Cook, D.S. Malkus, M.E. Plesha, R.J. Witt, *Concepts and Applications of Finite Element Analysis*, 4th ed., Wiley, 2002.
5. M.M. Khan, “AER626 Design Project: Bridge Design,” Department of Aerospace Engineering, Ryerson University, Toronto, ON, Canada, Jan. 1, 2018.

Appendix: SolidWorks Assembly Drawing

The following drawing shows the inverted T-stringer assembly modelled in SolidWorks, including the isometric view and cross-sectional dimensions used as the basis for the ANSYS finite element model.

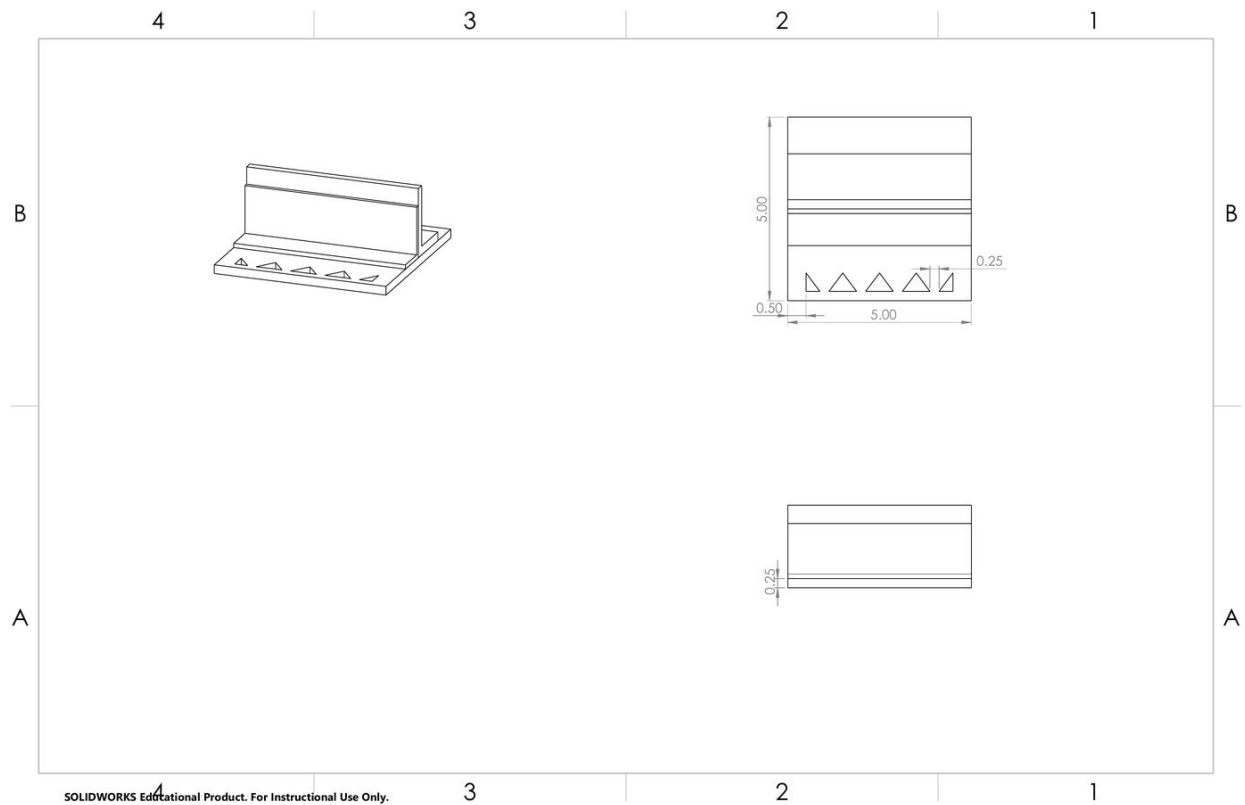


Figure 24: SolidWorks assembly drawing showing the inverted T-stringer geometry and dimensions.